

Christmas Math Crossword

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ONE ACROSS

This holiday season, the Grinch is up to no good once again; he is trying to steal as many presents as he can from the households of Whoville. Originally, his plan was to steal 1 present from each of the 250 houses located in Whoville. However, he is unsure whether there are any alternative plans that will allow him to steal more presents. If the Grinch has time to steal an additional present per household for every 10 households he skips, how many households should the Grinch ransack to maximize the amount of presents that he steals? For the sake of simplicity, assume that every household has a very large number of presents.

ONE DOWN

Santa is delivering presents to a street where the house numbers encompass every whole number from 1 - 575. As a part of his prescribed weight loss program, he decides to skip consuming the cookies and milk at every house which contains a 9 in their house number. Find the number of houses which he skips.

TWO DOWN

During the annual Winter Solstice, the Earth is tilted furthest away from the Sun, resulting in the least sunlight out of any day in the year. Assume that the Earth's axis of rotation is tilted 23.5 degrees away from the sun, as shown in figure 1. How long is the shadow of a 8.38 meter pole standing at the equator at noon, during the Winter Solstice in centimeters? For this problem, assume that there are no external factors beside the ones mentioned.

THREE DOWN

Santa has a total of N elves; however, he has forgotten what this number N really is. The number N satisfies the following properties:

- N is a four digit whole number
- The square of the value formed by summing the two digit number formed by the first two digits of N with the two digit number formed by the last two digits of N is N itself.
- N is the smallest number which satisfies the properties listed above

Find the sum of the first 8 digits of N^3 .

FOUR ACROSS

Mrs. Claus is making a frosted cookie that looks like the image shown on the right in figure 2. Given that the area of the green frosting rounded to the nearest whole number is $x \text{ cm}^2$, find $\lceil x \rceil$.

FIVE ACROSS

To manufacture toys, Santa's elves use a machine powered by cubic monic polynomial $P(x)$. Unfortunately, they have forgotten the password to this machine - $P(0)$. Given that $P(2) = P(P(2)) = P(P(P(2) + P(2))) = 8$, help save Christmas for the millions of little boys and girls around the world by finding the password.

1.	2.	3.
4.		
5.		



Figure 1

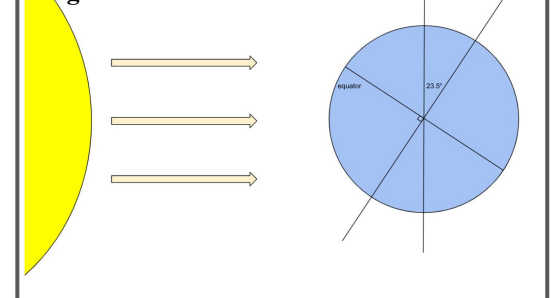


Figure 2

